W-C-430b
February 7, 1964

SUPERSEDING
Int. Fed. Spec. W-C-430a (GSA-FSS)
October 27, 1960

FEDERAL SPECIFICATION

CLEANER, VACUUM, (ELECTRIC, PORTABLE, FURNACE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The vacuum cleaners covered by this specification are of a heavy duty type for cleaning furnaces and boilers.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Type and sizes.* The vacuum cleaners covered by this specification shall be of one type and of the following sizes as specified (see 6.2):

Size 1—3/4 horsepower, minimum.
Size 2—1-1/2 horsepower, minimum

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

J-C-580—Cord, Flexible, and Wire, Fixture (Electrical, 0 to 600 Volt Service).
W-C-596—Connector, Plug, Electrical; Connector, Receptacle, Electrical; Plate, Wall, Electrical.
W-S-890—Switches; Snap, Miscellaneous.
CC-M-636—Motor, Alternating Current (Fractional Horsepower).
CC-M-641—Motor, Alternating Current (Integral Horsepower).
PPP-B-636—Box, Fiberboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking, for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

FSC 7910

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-P-116—Preservation, Methods of.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 — Quality Assurance Terms and Definitions.
MIL-STD-129—Marking for Shipment and Storage.
MIL-STD-130—Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Standards Association Standard:

C1-1962—National Electrical Code.

(Copies may be obtained from the American Standards Association, 10 East 40th Street, New York 16, New York.)

Underwriters' Laboratories, Inc. Standard:

No. 73—Motor Operated Appliances.

(Copies may be obtained from the Underwriters' Laboratories, Inc., 161 Sixth Avenue, New York 13, New York; 207 East Ohio Street, Chicago 11, Illinois; or 1655 Scott Lane, Santa Clara, California.)

3. REQUIREMENTS

3.1 Certification.

3.1.1 *Conformance to standards of Underwriters' Laboratories, Inc.* Each bidder shall submit to the contracting agency proof that the vacuum cleaner he proposes to supply under this specification conforms to the requirements of the Underwriters' Laboratories, Inc. Standard No. 73, Motor Operated Appliances. The label, or listing with re-examination, of the Underwriters' Laboratories, Inc., may be accepted as evidence that the vacuum cleaner conforms to this requirement.

3.1.2 *Data from other laboratories.* In lieu of label, or listing with re-examination, the bidder may submit independent proof, satisfactory to the contracting agency, that the vacuum cleaner conforms to the applicable requirements of the published standards including methods of tests of the Underwriters' Laboratories, Inc. Standard No. 73.

3.1.3 *Compliance by contractor.* Compliance with Underwriters' Laboratories, Inc., requirements regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his vacuum cleaners.

3.2 Materials. The materials used shall be suitable for the purpose intended and shall be free from defects affecting the appearance or serviceability of the cleaners.

3.3 Design and construction. The vacuum cleaner shall consist of a vacuum producer, dirt separator, and the attachments and tools specified (see 3.10). If the separator tank is connected to the vacuum producer by a hose, the hose shall be of suitable length and shall otherwise comply with the requirements of 3.10.1 and 3.10.1.1. Cleaners shall be rugged, reliable in operation, and constructed so as to be readily disassembled for inspection, adjustment, or replacement of parts. Each part of the cleaners shall be interchangeable with the similar part of any other cleaners of the same model and size made by the same manufacturer. Each cleaner shall be new and shall be the current product of the manufacturer.

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Each cleaner shall be fully lubricated and ready to operate.

3.4 **Body.** The entire exterior surface of the vacuum cleaner body shall be smooth. Ferrous metal parts, except corrosion resistant metal, shall be either zinc coated, suitably painted, or enameled.

3.5 **Vacuum producer.** The vacuum producer shall be enclosed and be of rugged construction and shall function without objectional noise or vibration. Surfaces of the impeller and air passages shall be constructed so that dirt will not build up on these surfaces to the extent of significantly affecting the flow of air. Means shall be provided to prevent metallic and heavy debris from entering the impeller case of the cleaner.

3.5.1 *Reversibility.* The vacuum cleaners shall be equipped for air blowing for the purpose of air sweeping. Changing from vacuum operation to air blowing operation and back to vacuum operation shall be accomplished without disassembling any portion of the cleaner and without the use of tools.

3.6 **Electric motor.** The electric motors for size 1 vacuum cleaners shall be in accordance with CC-M-636. The electric motors for size 2 vacuum cleaners shall be in accordance with the applicable provisions of CC-M-641. All motors shall be new and of the current production of the manufacturer. The motor shall be capable of developing the specified horsepower without the use of air from the vacuum producer for cooling purposes. Motors shall be classified as follows:

Type—universal.
Voltage—115 volts, unless otherwise specified (see 6.2).
Horsepower—as specified (see 3.11.1.1 and 3.11.2.1).
Speed—as specified (see 3.11.1.1 and 3.11.2.1).
Enclosure—protected or totally enclosed.
Duty—continuous.
Ambient temperature of reference—plus 40° C.

3.7 **Motor starter or switch.** A suitable snap switch for starting and stopping the motor shall be mounted on the cleaner. The rating of the switch shall be not less than the rating of the motor. The switch shall comply with the applicable requirements of W-S-890. Sufficient steps of resistance shall be provided where necessary for gradual acceleration of the motor up to normal speed.

3.8 **Cord assembly.** The electric cord assembly shall consist of a suitable flexible cord conforming to the applicable requirements of J-C-580 and a plug connector conforming to the applicable requirements of W-C-596. Unless otherwise specified (see 6.2), a 3 conductor jacketed cord, color coded (black, white, and green), of a construction equal to or exceeding type SO or ST cords of J-C-580 shall be provided with each cleaner. The conductors shall be of an American Wire Gage (AWG) size sufficient to carry 125 percent of the rated circuit current continuously, but in no case shall the conductors be smaller than AWG 18. One end of the green conductor shall be grounded to the cleaner in accordance with the applicable provisions of the American Standards Association (ASA) publication C1-1962 and the other end shall be connected to the ground terminal of the connector plug. Unless otherwise specified (see 6.2), the plug connector shall conform to the configuration requirements for style D plugs of W-C-596 and shall be either molded as an integral part of the cord or have a detachable cord grip connector fastened to the cord. Not less than 50 feet of cord shall be furnished with each cleaner.

3.9 **Dirt separator.** The dirt separator shall consist of a suitable tank for separating and receiving scale and heavy dirt particles and a dust bag for receiving fine

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dust. The vacuum producer may be attached directly to the tank or the tank may be separate from the vacuum producer and connected with a hose as specified in 3.3. The dirt shall be separated by the commercially designated "dry method."

3.9.1 *Tank.* The tank shall be made of steel and be of rugged construction and be dust tight. The tank shall have a capacity of not less than 1-1/2 cubic feet and shall be equipped with one or more carrying handles. All scale and heavy dirt particles shall be separated in the tank from the intake air before the air passes through the vacuum producer.

3.9.2 *Dust bag.* The dust bag shall be made of cloth suitable for retaining the fine dust and filtering the exhaust air passing through the bag. The dust bag may be either contained within the heavy dirt tank or mounted externally and connected to the vacuum producer exhaust at the option of the contractor. The dust bag shall be readily removable and easily cleaned. Externally mounted bags shall have an area of not less than 13 square feet and the joint between the bag and the vacuum producer shall be dust tight.

3.10 **Attachments and tools.** Unless otherwise specified (see 6.2), each vacuum cleaner shall be furnished with the following attachments and tools:

1 rubber connecting hose.
1 flexible steel hose.
1 tubular steel handle.
1 suction hand tool.
1 suction furnace tool.
1 1-1/2 inch round wire brush.
1 3 inch round wire brush.
1 15 inch by 2 inch wire brush.
1 flexible wire handle.

3.10.1 *Rubber connecting hose.* The hose for connecting the vacuum cleaner tools to the primary separator shall be not less than 12 feet long for the size 1 vacuum cleaner and not less than 25 feet long for the size 2 vacuum cleaner. The hose shall have a rubber lining and be covered by a durable rubber impregnated material. The hose shall be reinforced by a helix of spring steel wire embedded in the hose wall and extending the entire length of the hose. The hose shall be not less than 1-1/2 inches inside diameter and the weight of a 25 foot length with couplings shall not exceed 14 pounds.

3.10.1.1 *Rubber hose strength.* The hose shall be capable of withstanding a compressive load of 200 pounds applied externally and parallel to the longitudinal axis of the hose without causing a rupture of the hose and permanent diametrical deformation greater than 1/16 inch.

3.10.2 *Flexible steel hose.* The flexible steel hose shall be not less than 6 feet long and not less than 1 inch inside diameter. The hose shall be asbestos packed and sufficiently flexible for cleaning flues, back of furnaces, and areas difficult of access. The hose shall be fitted with a coupling on one end for attaching to the rubber connecting hose (see 3.10.1) and shall be open at the other end.

3.10.3 *Tubular steel handle.* The tubular steel handle shall be a straight steel tube not less than 46 inches long and not less than 1-1/4 inches inside diameter. One end of the tube shall be fitted with a bushing or other means for attaching to the rubber connecting hose and the other end shall be suitable for attaching the suction furnace tool.

3.10.4 *Suction hand tool.* The suction hand tool shall be made of aluminum, iron, or steel and shall be suitable for cleaning flat surfaces by means of suction. Means shall be provided for attaching the tool to the rubber connecting hose specified in 3.10.1. The overall width of the tool shall be not less than 5 inches nor more than 6 inches. The length of the vacuum slot in the tool shall be not less than 4-1/2 inches and the width not less than 1/4 inch.

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3.10.5 *Suction furnace tool.* The suction furnace tool shall be made of iron or steel. The tool shall be provided with a row of spring steel wire bristles or a steel scraper for loosening soot and a slot for the suction to carry the soot away. The bristles shall be not less than 0.008 inch in diameter and shall be approximately 1/2 inch in length. The bristles or the scraper shall be located in such position on the tool that all the loosened soot is drawn into the suction slot in the tool and not pushed ahead of the tool away from the suction zone of the tool. The width of the tool shall be 3 inches, plus or minus 1/4 inch. The width of the vacuum slot in the tool shall be not less than 1 inch. Means shall be provided for attaching the tool firmly to the tubular steel handle specified in 3.10.3.

3.10.6 1-1/2-*inch round wire brush.* The 1-1/2-inch round wire brush shall consist of spring steel wire bristles securely held by 4 twisted iron or steel wires to form a brush head 1-1/2 inches, plus or minus 1/4 inch in diameter and 4 inches, plus or minus 1/2 inch long. The bristle wire shall be not less than 0.008 inch in diameter. There shall be a sufficient number of bristles in the brush head to adequately remove scale and soot from boiler tubes and other surfaces within a furnace. The wire holding the bristles shall be not less than 0.120 inch in diameter. One end of the twisted wires shall be firmly secured in a fitting suitable for connecting to the detachable flexible wire handle specified in 3.10.9. In lieu of the fitting, the twisted wires may be extended not less than 3 feet beyond the bristle brush head to form a handle. The end of such a handle shall be formed into a loop approximately 1 inch in diameter.

3.10.7 3-*inch round wire brush.* The 3-inch round wire brush shall comply with the requirements specified in 3.10.6 except that the brush head shall be 3 inches, plus or minus 1/4 inch in diameter and 5 inches, plus or minus 1/2 inch long.

3.10.8 5-*by* 2-*inch wire brush.* The 5- by 2-inch wire brush shall consist of spring steel wire bristles securely held by a loop of four twisted iron or steel wires to form a brush head 5 inches, plus or minus 1/2 inch wide by 2 inches, plus or minus 1/2 thick by 4 inches, minimum, in length. The bristle wire shall be not less than 0.008 inch in diameter. There shall be a sufficient number of bristles in the brush head to adequately remove scale and soot from various surfaces in a furnace. The wire holding the bristles shall be not less than 0.105 inch in diameter. The ends of the twisted wires shall be firmly secured in a fitting suitable for connecting to the detachable flexible handle specified in 3.10.9. In lieu of the fitting, the twisted wires may be extended not less than 3 feet beyond the bristle brush head to form a handle. The end of such a handle shall be formed into a loop approximately 1 inch in diameter.

3.10.9 *Flexible wire handle.* (Not to be furnished if brushes have integral handles.) The handle shall be made of four or more twisted iron or steel wires and shall be not less than 4 feet long. One end shall be equipped with a fitting for attaching to the fitting on the brushes without integral handles specified in 3.10.6, 3.10.7, and 3.10.8. The other end of the handle shall be formed into a loop approximately 1 inch in diameter.

3.11 Performance.

3.11.1 *Size* 1, 3/4-*horsepower, minimum, vacuum cleaners.*

3.11.1.1 *Motor horsepower and speed.* The electric motor for size 1 vacuum cleaners shall be rated at not less than 3/4 horsepower. The full load speed of the motor shall be not more than 9000 revolutions per minute (r.p.m.).

3.11.1.2 *Suction.* The size 1 vacuum cleaner shall develop a suction at the end of a 25 foot length of 1-1/2 inch inside diameter hose of not less than 35 inches of water (2.5

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inches of mercury) with a 5/8 inch diameter orifice and not less than 23 inches of water (1.7 inches of mercury) with a 7/8 inch diameter orifice.

3.11.2 *Size 2, 1-1/2-horsepower, minimum, vacuum cleaners.*

3.11.2.1 *Motor horsepower and speed.* The electric motor for size 2 vacuum cleaners shall be rated at not less than 1-1/2 horsepower. The full load speed of the motor shall be not more than 8000 r.p.m.

3.11.2.2 *Suction.* The size 2 vacuum cleaner shall develop a suction at the end of a 25 foot length of 1-1/2 inch inside diameter hose of not less than 48 inches of water (3.5 inches of mercury) with a 5/8 inch diameter orifice and not less than 34 inches of water (2.5 inches of mercury) with a 7/8 inch diameter orifice.

3.12 **Dielectric strength.** The vacuum cleaner shall be capable of withstanding without breakdown, under the conditions specified in 4.4.9, the application of the given test potential between live metal parts and dead metal parts.

3.13 **Handbook of instructions.** One copy of the manufacturer's handbook of instructions for operation and maintenance and a list of repair parts shall be furnished with each vacuum cleaner.

3.14 **Identification marking.**

3.14.1 *Civil agencies.* Each vacuum cleaner shall be plainly and permanently marked with the manufacturer's designation for the type, model and size and with either his name or trademark of such known character that the source of manufacture may be readily determined. Complete data for the motor including the voltage and the horsepower output shall be given on the motor nameplate.

3.14.2 *Military agencies.* When specified (see 6.2), vacuum cleaners purchased for military use shall be marked in accordance with MIL-STD-130.

3.15 **Workmanship.** Vacuum cleaners shall be constructed, assembled, and finished in a thoroughly workmanlike manner to assure procurement of good quality equipment of a neat general appearance.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Inspection for acceptance.** Inspection shall be made in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter. The inspection terms and definitions shall be interpreted in accordance with MIL-STD-109.

4.2.1 *Inspection of components and materials.* In accordance with 4.1, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the specified requirements of referenced subsidiary specifications and standards, or, if none, in accordance with this specification.

4.3 **End product inspection.**

4.3.1 *Visual examination.* Examination shall be made of a completely assembled end item for defects in finish, material, design, construction, workmanship, and marking. Defects shall be classified as outlined below. The inspection level shall be level II with an

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6.3 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Vacuum cleaners, not otherwise indexed by name, with complement of attachments (specify weights of each).
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Vacuum cleaners, not otherwise indexed, with complement of attachments.
Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

6.4 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine conformance to the requirements of this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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AQL of 4.0 for major defects and 6.5 for total defects, expressed in defects per hundred units.

Examine	Defect	Major	Minor
Materials	Component not fabricated of the specified material.		X
Design	Any characteristic not in accordance with the specified requirements.	X	
Construction and workmanship general (Applicable to all components and assemblies)	Part missing or not specified type or size.	X	
	Fractured, split, bowed, malformed, or otherwise impaired affecting serviceability.	X	
	Defective or otherwise damaged but not affecting serviceability.		X
	Misplaced, loose or not in proper alignment affecting serviceability.	X	
	Sharp burr, sliver, or splinter that may be injurious to personnel or may affect serviceability.	X	
Electrical assembly	Wiring cut, abraded, not properly joined, loose at terminal.	X	
	Adequate slack not provided for wiring to relieve strain, or excessive insulation stripped from wiring.		X
Lubrication (where applicable)	Not lubricated where required.	X	
	Insufficiently lubricated.		X
Assembly	Not properly assembled or secured.		X
Attachments and tools	Not furnished when required.	X	
Marking identification	Missing, incomplete, not legible.		X

4.3.2 *Dimensional examination.* Examination shall be made of the end item to determine compliance with dimensional requirements. Any dimension that exceeds the specified tolerance shall be classified a defect. The inspection level shall be L-4 with an AQL of 6.5 defects per hundred units.

4.3.3 *Testing of the end products.* The contractor shall test the end item in accordance with 4.4.1 through 4.4.9 using inspection level L-8 with an AQL of 4.0 defects per hundred units.

4.4 **Tests.** Samples of end products shall be selected for testing in accordance with 4.3.3.

4.4.1 *Continuous run test.* Each sample vacuum cleaner selected for testing shall be subjected to a continuous run test prior to conducting the tests specified herein. The cleaner shall be operated in the condition received continuously for 10 hours with a 3/4-inch-diameter orifice in the end of the suction hose attached to the cleaner. The cleaner shall not be lubricated during this run and shall have adequate lubricant in the bearings at the end of the run. At the end of this run, motors shall operate with practically sparkless commutation which shall be understood to mean an entire absence of sparking other than fine pinpoint blue sparks along the edges of the brushes. Heavy sparks or yellow sparks shall be considered as failure to comply with this requirement. The vacuum producer shall be observed for excessive noise and vibration during this test to determine conformance to 3.5.

4.4.2 *Reversibility.* The vacuum cleaner shall be operated to blow air to determine conformance to 3.5.1.

4.4.3 *Motor and switch tests.* The motor and switch of each sample vacuum cleaner shall be tested in accordance with the applicable provisions of Underwriters' Laboratories, Inc., Standard No. 73 to determine compliance with the requirements of this specification.

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4.4.4 *Rubber hose strength.* The hose shall be subjected to a compressive load test to determine conformance to 3.10.1.1. A compressive load of 200 pounds shall be applied to any portion of the hose by means of two parallel surfaces having a length of 3 inches parallel to the axis of the hose. The load shall be applied for 2 minutes and then released.

4.4.5 *Suction tool test.* The suction hand tool and the suction furnace tool specified in 3.10.4 and 3.10.5 respectively, shall be tested for furnace or equivalent cleaning to determine that the tools are satisfactory for their intended use.

4.4.6 *Wire brush test.* The wire brushes specified in 3.10.6, 3.10.7 and 3.10.8 shall be tested for furnace or equivalent cleaning to determine that the brushes are satisfactory for their intended use.

4.4.7 *Exposed dust bag test.* During the suction tool test of 4.4.5 the exposed dust bag specified in 3.9.2 shall be tested for dust retention and air filtering. With the bag attached to the vacuum cleaner the cleaner shall be operated for 2 minutes picking up soot or an equivalent material. At the end of the 2 minute picking-up operation a thin piece of damp muslin about 8 inches square shall be placed on the central portion of the upper surface of the dust bag and kept there while the cleaner is operating to suck fresh air for a period of 20 minutes. At the end of the 20 minute run the muslin shall not be appreciably soiled.

4.4.8 *Suction tests.* Sample vacuum cleaners shall be tested for suction to determine conformance to 3.11.1.2 for size 1 and 3.11.2.2 for size 2 vacuum cleaners. The tests shall be conducted at an ambient temperature of approximately 68° F. and barometric pressure of approximately 29.00 to 30.00 inches of mercury. Suction requirements shall be met without causing the motor to exceed its allowable maximum operating temperature.

4.4.8.1 *Test procedure.* The suction test chamber shall be as shown in figure 1. A 25-foot-length of hose shall be attached to the test chamber and the hose laid out straight so as to be approximately in line with the intake connection of the vacuum cleaner. The specified size of sharp edged orifice shall be installed in the chamber. The vertical distance between the water levels in the two columns of a manometer connected to the test chamber shall be measured to the nearest 0.10 inch for determining the suction of the vacuum cleaner. All connections to the test chamber and vacuum cleaner shall be air tight. Uniform running conditions shall be established by a suitable warm up period of operation. Performance data obtained by extrapolation of actual test data shall not be given.

4.4.9 *Dielectric strength.* Sample vacuum cleaners shall be tested for dielectric strength to determine conformance to 3.12. A 60-cycle alternating potential of 1000 volts, plus twice the rated voltage of the vacuum cleaner motor, shall be applied between live metal parts and dead metal parts with the motor at the maximum operating temperature reached in normal use. The test shall be conducted with a suitable transformer, the output voltage of which can be regulated. Starting at zero, the applied potential is to be gradually increased until the required test voltage is reached and is to be held at that value for one minute.

4.5 Examination of preparation for delivery. An examination shall be made to determine that the requirements of section 5 for preservation, packaging, packing, and marking are complied with. Defects shall be scored as specified below. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The inspection level shall be L-7 with an AQL of 6.5 defects per hundred units.

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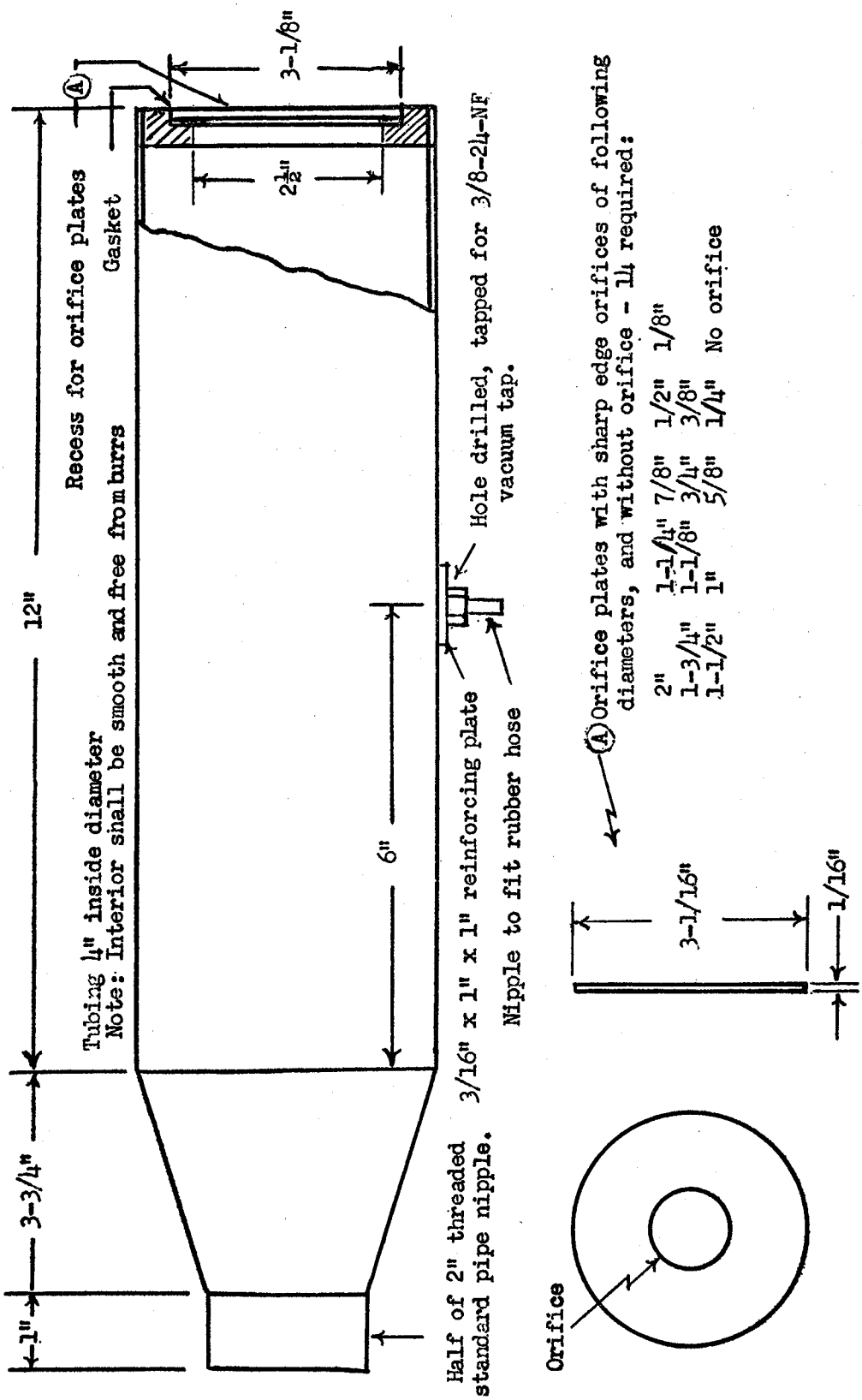


Figure 1. - Testing chamber for vacuum cleaners.

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Examination	Defect
Markings, exterior and interior	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Preservation	Preservation improperly applied or missing.
Materials	Component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components such as incomplete closure of case liners and container flaps, loose strapping, inadequate stapling; bulging and distortion of containers; blocking and cushioning inadequate, improper, or missing; bolts improperly affixed.
Weight	Gross weight exceeds requirements.

5. PREPARATION FOR DELIVERY

(For Civil agency procurement, Fed. Std. No. 102 should be referred to for definitions and applications of the various levels of preservation, packaging, and packing.)

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.* The vacuum cleaners, attachments, accessories, and repair parts shall be preserved and packaged in accordance with method IC-5 of MIL-P-116. Instruction publications and parts lists shall be packaged in accordance with method IC-3 of MIL-P-116.

5.1.2 *Level C.* The vacuum cleaners, attachments, accessories, repair parts, and instruction publications shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* The packaged equipment shall be packed in close fitting boxes conforming to PPP-B-636, class 2, weather resistant.

5.2.2 *Level B.* The packaged equipment shall be packed in boxes conforming to PPP-B-636, class 1, domestic service.

5.2.3 *Level C.* The packaged equipment shall be packed to insure carrier acceptance and safe delivery to destination in compliance with rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order (see 6.2), interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order (see 6.2), interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The vacuum cleaners covered by this specification are intended for use in removing dirt, soot, and scale from furnaces, boiler tubes, and burners. They are intended also for air sweeping when used as a blower.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Size required (see 1.2.1).
(*c*) Motor voltage if different from 3.6.
(*d*) Electric cord if different from 3.8.
(*e*) Plug connector if different from 3.8.
(*f*) Attachments and tools furnished if different from 3.10.
(*g*) When vacuum cleaners shall be marked in accordance with MIL-STD-130 (see 3.14.2).
(*h*) Level of preservation and packaging required (see 5.1).
(*i*) Level of packing required (see 5.2).
(*j*) Special marking, when required (see 5.3.1 and 5.3.2).

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